# Aarush Ram Anandh

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Hiring Team
Google

**Application: Student Researcher**

Dear Hiring Team,

I am applying to the Student Researcher program with a focus on machine learning, computer vision and optimization. I am completing a B.Sc. (Hons.) in Computer Science and Mathematics at Ashoka University, with graduation expected in 2028. My work combines mathematical investigation with implementations and experiments on image, audio and physiological data.

At the Constrained Image-Synthesis Lab, I improved colour fidelity in latent diffusion models, achieving $\Delta E_{00} < 1$ and approximately four times better colour accuracy than evaluated open-source baselines. This work is reported in the public preprint *Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling* (arXiv:2601.17259). I am also developing Controlled Generalised Sliced Wasserstein, a controllable optimal-transport framework with a paper submitted to JMLR. That work includes C++ computational methods, Lean specifications and proofs, and an ongoing hyperspectral class-separability experiment.

In a two-person reinforcement learning coursework project, we built a masked-action environment for Pokemon battles and evaluated PPO across two team variants and five seeds per variant. PPO reached 99.7–100% wins against the fixed heuristic over 1,020 final battles per run; the study found no significant learning-speed difference between the team variants. At the Koita Centre for Digital Health, I contributed to a wearable-data pipeline achieving approximately 70% sensitivity for asthma exacerbation detection in study evaluation. The work includes smoothing and change-point features for 191 patients, with patient-wise splits and bootstrap intervals. At Avartan Labs, I raised rare-class correct detection from 40% to 80% in an RT-DETR detection and tracking pipeline covering 18 material classes.

I would welcome a research placement where I can contribute to method development, careful evaluation and implementation alongside experienced researchers. The program's range of exploratory projects is particularly appealing because my strongest interests lie where mathematical structure meets empirical model behaviour. Thank you for considering my application.

Yours sincerely,

Aarush Ram Anandh